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ORIGINAL ARTICLE

Mitigating GPS spoofing in AVS: SHA and RSA algorithms with proteus simulation for real-time detection

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Abstract
Smart car Autonomous Vehicular Systems (AVS) face significant threats from GPS spoofing, which can dangerously manipulate a vehicle's perceived location. This paper presents a robust and enhanced mitigation framework that combines machine learning-based detection, cryptographic safeguards, and real-time alert systems. The proposed methodology begins with system initialization and GPS signal reception, followed by a data consistency check that flags suspicious signals. Statistical analysis and supervised machine learning, developed in prior work, identify spoofed signals based on key features such as PDOP, HDOP, and VDOP, with PDOP emerging as a primary indicator. The Extra Trees Regressor model demonstrates superior performance in handling these complex interactions, with visualizations such as heatmaps and scatter plots offering deeper insights. To further enhance reliability, an MPU6050 accelerometer-gyroscope module is integrated to continuously monitor real-time motion and orientation. This enables inertial measurements to cross-verify GPS data, especially in spoofing scenarios or environments with signal obstructions. When spoofing is confirmed, the system triggers LED indicators and sends alerts via GSM, while normal data is hashed using SHA-256, encrypted with RSA, and securely transmitted. This comprehensive approach ensures high integrity, real-time spoofing detection, and system resilience, significantly enhancing safety and operational reliability in AVS and other GPS-dependent applications.

Keywords GPS spoofing detection · Autonomous vehicular systems (AVS) · Machine learning · Cryptographic security · Inertial measurement (MPU6050)

1 Introduction

1.1 An overview of GPS spoofing attacks and how they affect AVS

Autonomous Vehicular Systems (AVS) in smart cars heavily rely on accurate GPS data for navigation, positioning, and safety-critical decisions. However, GPS spoofing attacks, where fraudulent GPS signals are transmitted to deceive the vehicle's navigation system, pose a significant threat to AVS. These attacks can manipulate the perceived location of the vehicle, leading to hazardous situations such as navigation errors, route deviations, or even collisions. The increasing prevalence of such attacks underscores the urgent need for robust countermeasures to guarantee GPS data security and integrity in AVS.

1.2 Motivation for using SHA and RSA algorithms for mitigation

To tackle the difficulties caused by GPS spoofing, this study investigates the incorporation of cryptographic methods, specifically SHA-256 and RSA algorithms, into the GPS data processing framework of AVS. The SHA-256 algorithm provides a secure method for hashing GPS data, generating a distinct hash value for every data set to guarantee its integrity. It is very difficult for attackers to change GPS data, covertly thanks to this hashing process. In addition, the

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RSA algorithm provides a dependable way to encrypt the hashed data, enabling safe authentication and transmission. The combined use of these cryptographic techniques provides a robust defence against GPS spoofing by safeguarding data integrity and authenticity.

1.3 Objectives of the paper

This paper's main goal is to provide a thorough mitigation strategy for GPS spoofing attacks in AVS by utilizing RSA encryption and SHA-256 hashing.

The proposed system aims to:

1. Ensure Data Integrity and Security: Implement SHA-256 hashing to detect any alterations in the GPS data and RSA encryption to secure the transmission of this data.
2. Enable Real-Time Detection and Response: Develop a real-time detection mechanism using Proteus simulation, where spoofing attempts are promptly identified and indicated through an LED indicator, with notifications sent via a GSM module.
3. Provide a Framework for Effective Debugging and Response Handling: Integrate interfaces for serial communication, debugging, and response handling to diagnose and resolve system issues efficiently.

By achieving these objectives, this paper seeks to enhance the resilience of AVS against GPS spoofing attacks, thereby enhancing the general dependability and safety of driverless cars. A methodical strategy for preventing spoofing attacks using GPS in autonomous vehicles using SHA and RSA algorithms, including simulation in Proteus and notification to GSM, is proposed.

1.4 Contribution of the paper

The paper offers a strong framework to counter GPS spoofing attacks, which helps to improve the security of autonomous vehicular systems (AVS). It combines real-time detection mechanisms with cryptographic techniques, like SHA-256 hashing for data integrity and RSA encryption for safe transmission. Through Proteus simulation, the system detects spoofing attempts and alerts users via GSM and LED. The added MPU6050 sensor boosts reliability by verifying GPS data with motion readings, ensuring accurate spoof detection even in compromised environments. By combining secure GPS data processing with real-time response capabilities, this work offers a practical solution to safeguard AVS against location manipulation, ensuring reliability and safety in smart cars.

1.5 Organization of the paper

The remainder of this paper is organized as follows: Sect. 2 provides a comprehensive literature review covering prior work on GPS spoofing detection in autonomous and unmanned aerial vehicles, innovations in low-cost control systems and safety enhancements for modern vehicles, and advanced authentication and key management strategies for secure vehicular communication. Section 3 outlines the methodology, detailing the proposed framework for detecting spoofing threats, the integration of SHA and RSA cryptographic techniques for ensuring GPS data integrity and authenticity, the design of a secure GPS data processing model including debugging interfaces, the simulation environment, and the procedure for collecting live GPS data using a GPS tracker. Section 4 presents the results, highlighting the detection accuracy of valid versus spoofed GPS data and analysing the overall effectiveness of the proposed approach. Finally, Sect. 5 concludes the study and outlines future research directions.

2 Literature review

2.1 GPS spoofing detection and mitigation techniques in autonomous and unmanned aerial vehicles

In their paper, Y. Dang and (A) Karakoc used maps and techniques of ML to detect and prevent attacks like spoofing of GPS against Unmanned Aerial Vehicles (Dang et al. 2023). (B) Pardhasaradhi, P. Srihari, and P. Aparna propose a track filter with a robust positioning algorithm that uses both real and fake GPS signals to enhance positioning accuracy and counter spoofing effects (Pardhasaradhi et al. 2021). To detect spoofing attacks, B. Pardhasaradhi and L. R. Cenkeramaddi proposed a fusion-based approach including two methods to identify the spoofing attack (Pardhasaradhi and Cenkeramaddi 2022). The paper by R. T. Ioannides, T. Pany, and G. Gibbons describes the basic ideas of satellite navigation and analyses the consequences (Ioannides et al. 2016). To prevent spoofing attacks, Pardhasaradhi mainly applied sparse estimation theory (Pardhasaradhi et al. 2022).

N. Carson, S. M. Martin, J. Starling, and D. M. Bevly proposed the detection of spoofing which is based on a control system that provides sharing of data for vehicle communication (Carson et al. 2016). In their paper, A. Oracevic and A. Salman looked at common cyber threats and possible defences (Oracevic and Salman 2024). A study by H. Mukkath, A. Altaweel, I. Kamel, and Z. Al Aghbari contrasts and compares several detection methods for GPS spoofing attacks (Mukkath et al. 2023). To investigate the

effectiveness of a spoofing detection algorithm, C. Jiang and S. Chen used a covariance matrix (Jiang et al. 2018). N. Piperigkos, A. S. Lalos, and C. Anagnostopoulos explored the effects of cyberattacks like GPS spoofing on autonomous vehicles and emphasized the creation of effective, centralized systems for detecting and countering such threats, ensuring location accuracy and security oversight across the entire vehicle fleet (Piperigkos et al. 2023).

With a focus on the elements of cyber-security in automotive settings, C. Giannì and R. Giordano's study addressed GNSS Threat Identification and Mitigation (Giannì et al. 2023). In a paper, V. Tiwari and A. Agrawal suggest a subspace projection-based mitigation strategy to lessen GPS technology's vulnerability to spoofing and jamming attacks (Tiwari et al. 2023). M. (A) Hoque and R. Hasan assessed the threat landscape and examined security challenges in autonomous driving systems using the STRIDE threat modelling framework (Hoque and Hasan 2022). D. K. S. Medel, (B) A. M. Perez, P. V. Alpaño, and J. R. Pedrasa developed a blockchain-based ITS using smart contracts and PBFT, simulating data integrity attacks like GPS spoofing and sensor deception. The system reduced attack effects and improved security (Medel et al. 2021).

N. Vagle and A. Broumandan's study focuses on integrating multi-antenna GNSS with an inertial navigation system (INS) and odometer to strengthen navigation solutions' privacy, security, and resilience (Vagle et al. 2018). M. Brachvogel, M. Niestroj, and M. Meurer propose a method for calibrating a distributed subarray, estimating differential delays and frequency-dependent transfer characteristics for wideband signals like GPS L5 and Galileo E5a (Brachvogel et al. 2023). Poornima B., & Kumari, L. S. (2024) propose a machine learning-based approach to enhance GPS spoofing detection, improving the reliability and safety of autonomous vehicles (Poornima and Kumari 2025).

Table 1 provides a concise summary of key GPS spoofing detection and mitigation techniques specifically developed for autonomous and unmanned aerial vehicles, highlighting their methodologies, core technologies, and effectiveness in enhancing navigation security.

2.2 Innovative Low-Cost control systems and safety enhancements for modern vehicles

H. Shabani, M. R. M. Razif, and M. F. Muhsin developed an affordable smart car control system to detect and avoid road obstacles (Shabani et al. 2021). To design a communication system between the road and the vehicle for wireless speed alerting and control via XBEE, F. Y. Mohammed and A. A. Osman's study employed a two-transceiver module as part of its methodology (Mohammed and Osman 2017). An AT89C52 MCU-based steering light control system for cars was designed by Y. Gong and P. Zhang in their paper (Gong and Zhang 2023).

The Battery Management System (BMS) was created by R. S. Kumar, D. S. Anusiya Devi, and J. S. Linda Johnsana to guarantee the efficiency, security, and maintenance of the electric vehicle (EV) (Kumar et al. 2023). K. Ramanan, V. S., N. S., and M. H. Thaqfeen's research aims to develop an LPG leak detection and reaction system using a microcontroller (Ramanan et al. 2023). An Arduino board will use a pre-burned code to start emergency braking after a sonar

Table 1 Summary of key GPS spoofing detection and mitigation techniques in autonomous and unmanned aerial vehicles

Authors	Title	Main Focus / Technique	Application Area	Key Contribution
B. Pardhasaradhi et al.	Navigation in GPS spoofed environment using M-best positioning algorithm and data association	Track filter + robust positioning algorithm	General GPS	Combines real & fake GPS signals to improve accuracy
Pardhasaradhi	GPS spoofing detection and mitigation for drones using distributed radar tracking and fusion.	Sparse estimation theory	GPS spoofing	Spoofing prevention using sparse signal processing
N. Carson et al.	GPS spoofing detection and mitigation using Cooperative Adaptive Cruise Control system	Control system-based spoofing detection	Autonomous vehicles	Inter-vehicle data sharing for spoofing detection
N. Piperigkos et al.	Robust cooperative sparse representation solutions for detecting and mitigating spoofing attacks in autonomous vehicles	Centralized detection system	Autonomous vehicles	Fleet-wide spoofing countermeasures
V. Tiwari & A. Agrawal	Mitigation of jamming and spoofing attack on GNSS signals using subspace projection	Subspace projection method	GNSS spoofing/jamming	Robust mitigation of spoofing/jamming effects
D. K. S. Medel et al.	Mitigation of data integrity attacks using blockchain-based intelligent transportation system.	Blockchain-based ITS using smart contracts	ITS / spoofing mitigation	Integrity and security enhancement against GPS spoofing
N. Vagle & A. Broumandan	Multiantenna GNSS and inertial sensors/odometer coupling for robust vehicular navigation	Multi-antenna GNSS + INS + odometer integration	Navigation resilience	Enhancing GPS spoofing resilience via multi-sensor fusion

sensor detects obstacles in the system developed by M. S. Rezwan, M. A. Islam, M. M. Islam, and M. R. Hasan (Rezwan et al. 2018).

The Proteus Lab Center simulation that employed a battery model with energy recovered from a reversible motor based on functional criteria served as the basis for the study by N. Nistor, L. Baicu, and B. Dumitrascu (Nistor et al. 2022). Parida, P., Pradhan, C., Alzubi, J.A. et al. designed a model that uses ECDH key exchange, chaotic maps (Henon, Hopfield), and improved ElGamal encryption with digital signatures for secure image encryption. It ensures high security, robustness, and authenticity against various cryptographic attacks (Parida et al. 2023). Alzubi, O.A., Alzubi, J.A., Dorgham, O. *et* al proposed a methodology that uses Hermitian curves and algebraic geometric codes to develop the Hermitian-based cryptosystem (HBC) with strong encryption and error correction. It provides high security, large key sizes, and better error resilience, making it ideal for IoT applications (Alzubi et al. 2020).

Table 2 presents the Extended GPS Spoofing Summary, offering a detailed comparison of advanced detection strategies and mitigation frameworks, including the integration of cryptographic techniques and sensor fusion approaches, to bolster GPS reliability in autonomous systems.

2.3 Advanced authentication and key management strategies for secure communication in vehicular networks

Poornima, B., & Kumari, P. L. S. (2024) presented a dual-step approach for implementing Smart Autonomous Vehicular Systems (AVS) in cars, which involves designing the architecture using the Microsoft Threat Modelling tool and simulating smart cars with Engineering Software (Poornima and Kumari 2024). J. Zhang, L. Li, W. Liu, and X. Li (Zhang et al. 2022) introduced a one-way hash chain authentication method with a key update strategy. S. Kumar and V. Singh (Kumar and Singh 2021) reviewed algorithms ensuring secure network communication. C. Jeong and Y. Kim implemented the SHA-256 FPGA module for the IEEE 1609.2 VC security protocol (Jeong and Kim 2014). H. Vasudev and D. Das proposed an efficient authentication system for V2V communications in IoV (Vasudev and Das 2019). Q. Shi et al. (Shi et al. 2024) introduced QKBAKA, a quantum-key-based authentication scheme for IoV. N. V. Vighnesh and N. Kavita (Vighnesh et al. 2011) presented a sender authentication scheme to enhance VANET security.

Y. Xiaodong et al. proposed a VANET message authentication system using a trapdoor hash and proxy re-signature (Xiaodong et al. 2018). I. Ahmim et al. (Ahmim et al. 2022) analysed an IoV authentication and key agreement model. M. Malathi et al. (Malathi et al. 2012) proposed a multilevel authentication system for one-to-one group communication. H. Vasudev, V. Deshpande, D. Das, and S. K. Das developed a protocol that reduces computational overhead by establishing a secret key for secure communication between devices and servers using Raspberry Pi setups (Vasudev et al. 2020). They also proposed a lightweight authentication protocol for securing EVs in VANETs (Wang et al. 2021). P. Vijayakumar et al. (Vijayakumar et al. 2016) established a trusted authority to provide premium services via VANETs.

Table 3 provides an overview of advanced authentication mechanisms, highlighting key strategies such as cryptographic hashing, key management protocols, and lightweight security frameworks designed to ensure secure communication within vehicular networks.

Table 2 Extended GPS spoofing summary

Authors	Title	Technology/Methodology	Application	System Used
H. Shabani, M. R. M. Razif, and M. F. Muhsin	Affordable Smart Car Control System for Obstacle Detection and Avoidance	Obstacle detection and avoidance with smart sensors	Smart control system for vehicle obstacle avoidance	Smart car sensors
F. Y. Mohammed and A. A. Osman	Wireless Speed Alerting and Control via XBEE	XBEE communication system with a two-transceiver module	Wireless communication for vehicle speed monitoring and control	XBEE transceivers
Y. Gong and P. Zhang	Steering Light Control System Using AT89C52 MCU	AT89C52 Microcontroller	Car steering light automation	AT89C52 MCU
R. S. Kumar, D. S. Anusiya Devi, and J. S. Linda Johnsana	Battery Management System for Electric Vehicles	Battery Management System (BMS)	Efficiency and safety in EV battery systems	BMS for EVs
K. Ramanan, V. S., N. S., and M. H. Thaqfeen	LPG Leak Detection and Reaction System Using Microcontroller	Microcontroller-based LPG sensor system	Leak detection and automatic response for LPG leaks	Microcontroller system
M. S. Rezwan, M. A. Islam, M. M. Islam, and M. R. Hasan	Emergency Braking System Using Sonar Sensor and Arduino	Arduino, pre-burned code, sonar sensor	Autonomous emergency braking	Arduino board and sonar sensor
N. Nistor, L. Baicu, and B. Dumitrascu	Mitigating GPS Spoofing in AVS: SHA and RSA Algorithms with Proteus Simulation for Real-Time Detection	SHA and RSA Algorithms, Proteus Simulation	Real-time GPS spoofing detection in Autonomous Vehicles	Proteus Lab Center Simulation

Table 3 Advanced authentication summary

Authors	Title	Technology/Methodology	Application	System Used
Poornima, B., & Kumari, P. L. S.	Dual-step AVS Implementation with Microsoft Threat Modelling and Engineering Software	Microsoft Threat Modelling Tool, Engineering Simulation Software	Secure architecture design and simulation for Smart AVS	Smart AVS architecture and simulation platforms
J. Zhang, L. Li, W. Liu, and X. Li	One-way Hash Chain Authentication with Key Update Strategy	One-way Hash Chain, Key Update Strategy	Secure vehicular network communication	Vehicular network systems
C. Jeong and Y. Kim	SHA-256 FPGA Module for IEEE 1609.2 VC Security Protocol	SHA-256, FPGA, IEEE 1609.2 Protocol	Vehicular communication security	FPGA systems
H. Vasudev and D. Das	Efficient Authentication for V2V Communication in IoV	Authentication Protocol for IoV	Vehicle-to-Vehicle (V2V) communication in IoV	IoV network
Q. Shi et al.	QKBAKA: Quantum-Key-Based Authentication Scheme for IoV	Quantum Key Authentication	Authentication in Internet of Vehicles (IoV)	Quantum communication systems
M. Malathi et al.	Multilevel Authentication for One-to-One Group Communication	Multilevel Authentication System	Secure one-to-one communication in VANET	Group communication setup in VANET
H. Vasudev, V. Deshpande, D. Das, and S. K. Das	Secret Key Establishment Protocol Using Raspberry Pi for Secure Communication	Secret Key Protocol, Raspberry Pi	Lightweight secure communication in VANET using embedded systems	Raspberry Pi and embedded systems

3 Methodology of cryptographic GPS spoofing detection and mitigation for autonomous vehicles

3.1 Proposed framework for identifying spoofing threats

The Framework illustrated in Fig. 1, used to identify and analyze potential threats in autonomous vehicular systems (AVS) in smart cars, follows a four-step process for threat mitigation. First, it identifies the assets to be protected, such as the car, driver, and passengers. Next, it identifies potential threats to these assets, including spoofing, denial of service attacks, and data breaches. After identifying threats, it finds vulnerabilities that these threats could exploit, such as unsecured wireless networks, weak passwords, and outdated software. The proposed framework provides a comprehensive approach to identifying potential threats to the Automatic vehicular system in smart cars. By following this framework, car manufacturers and security professionals can take a proactive approach to security and help protect against potential attacks. Once the spoofing threats are identified through the Proposed Threat Classification Approach, it is essential to develop mitigation strategies to prevent or minimize the impact of such attacks.

Autonomous vehicle systems (AVS) are seriously threatened by GPS spoofing, which could lead to misnavigation and jeopardize passenger safety. Building on my previous research on machine learning-based GPS spoofing detection and smart autonomous vehicles (AVS), this paper offers a thorough methodology that uses statistical analysis and machine learning techniques to detect GPS spoofing (Poornima and Kumari 2025). Model building and evaluation are the two stages of the methodology. During the construction phase, a GPS dataset with both spoofed and non-spoofed signals is processed using PyCaret, with an emphasis on important features like PDOP, HDOP, and VDOP. Missing data handling, categorical variable encoding, and hyperparameter optimization across different classification models are examples of preprocessing steps. The dataset's analysis showed that PDOP is a significant indicator of spoofing, with spoofed signals typically displaying higher PDOP values.

The ability of the Extra Trees Regressor to handle intricate feature interactions made it stand out among the tested models, with VDOP turning out to be the most significant feature. Histograms, heatmaps, and scatter plots were among the visualizations that provided additional insight into feature distributions and how they relate to spoof signals. A strong framework for real-time GPS spoofing detection was produced by the integrated pipeline, which combined feature selection, data preprocessing, and thorough model evaluation.

With additional applicability in other GPS-reliant industries like maritime and aviation, these findings emphasize the significance of implementing sophisticated spoofing detection systems in AVS to enhance security and reliability. The insights and methodologies from this prior work have directly contributed to the current paper's process, where the focus shifts from mere detection to implementing effective spoofing mitigation strategies, thereby enhancing the overall resilience and operational safety of AVS.

3.2 Combining SHA and RSA to ensure the authenticity and integrity of GPS data

A cryptographic hash function called SHA-256 creates a fixed 256-bit hash value. It ensures data integrity by giving each input a unique output; even small changes result in

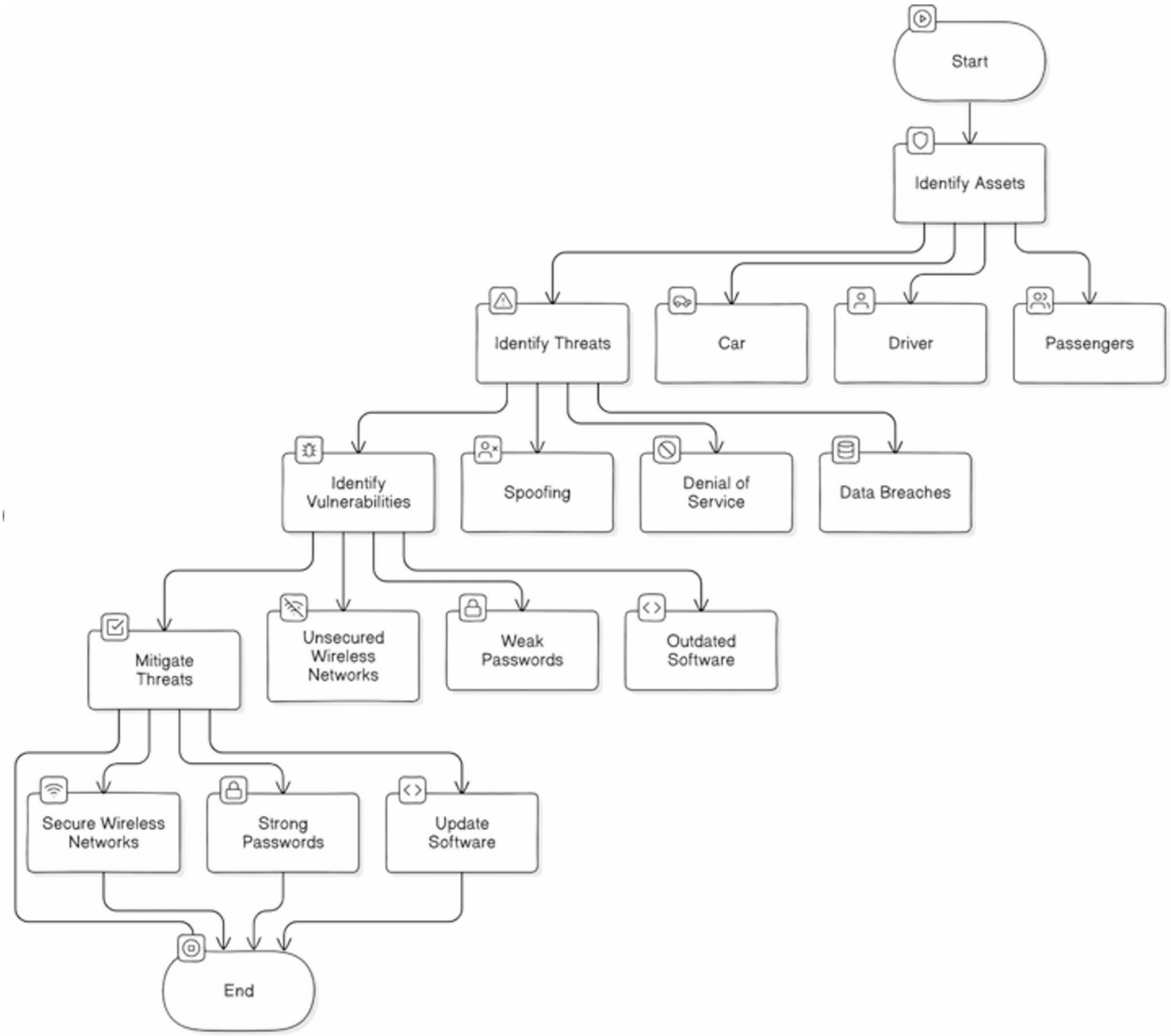


Fig. 1 Proposed Framework in Identifying Spoofing Threats

large hash variations. Since it is a one-way function, recovering the original input from the hash is not computationally feasible. A public key is used for encryption, and a private key is used for decryption in the RSA public-key cryptographic algorithm. RSA, which is widely used for digital signatures, guarantees data authenticity and integrity by enabling the verification of data signed with a private key using the matching public key. The suggested methodology combines the SHA-256 and RSA algorithms to enhance the security and integrity of GPS data in Autonomous Vehicular Systems (AVS) in smart cars.

The following steps outline the integration process:

1. Data Reception: The system initializes serial communication and sets up the GPS module to receive GPS signals.
2. Data Hashing (SHA-256): The received GPS data is processed and hashed using the SHA-256 algorithm. To ensure that any changes in the GPS data can be identified, the SHA-256 algorithm creates a distinct 256-bit hash value from the data.
3. Data Signing (RSA): The generated hash value is signed using the RSA private key. By encrypting the hash value with the private key, the RSA algorithm generates a digital signature that offers a means of confirming the data's legitimacy.
4. Data Transmission and Verification: The signed hash (digital signature) is sent to the GPS module or another

verification system for authentication. The verification system decrypts the signature using the RSA public key, then compares the generated hash with the hash of the data it received. An LED indicator is activated to confirm the authenticity of the data if the hashes match; if not, the LED stays off, suggesting possible spoofing.

3.3 Proposed model for secure GPS data processing system with cryptographic techniques and debugging interface

A secure GPS data processing system that uses cryptographic techniques and a debugging interface to guarantee data integrity and security is depicted in the diagram in Fig. 2. An initialization module starts the system by configuring everything and establishing serial communication with external devices.

The GPS module receives signals and data, which are then sent to the GPS Data Processing unit for processing into usable information for location tracking or other applications. To ensure its integrity and security, this processed data is hashed using SHA-256 to produce a fixed-size hash value. The hashed data is then encrypted using RSA keys, a public-key cryptographic algorithm that ensures secure data transmission. A response handling component manages and sends appropriate responses based on the processed GPS data, possibly triggering actions or alerts. An LED indicator provides visual status updates or alerts based on the system's processing results. Additionally, a debugging interface is included to aid in diagnosing and resolving system issues. Overall, the system is designed to securely process GPS data by integrating cryptographic techniques, while offering communication, response handling, and debugging capabilities.

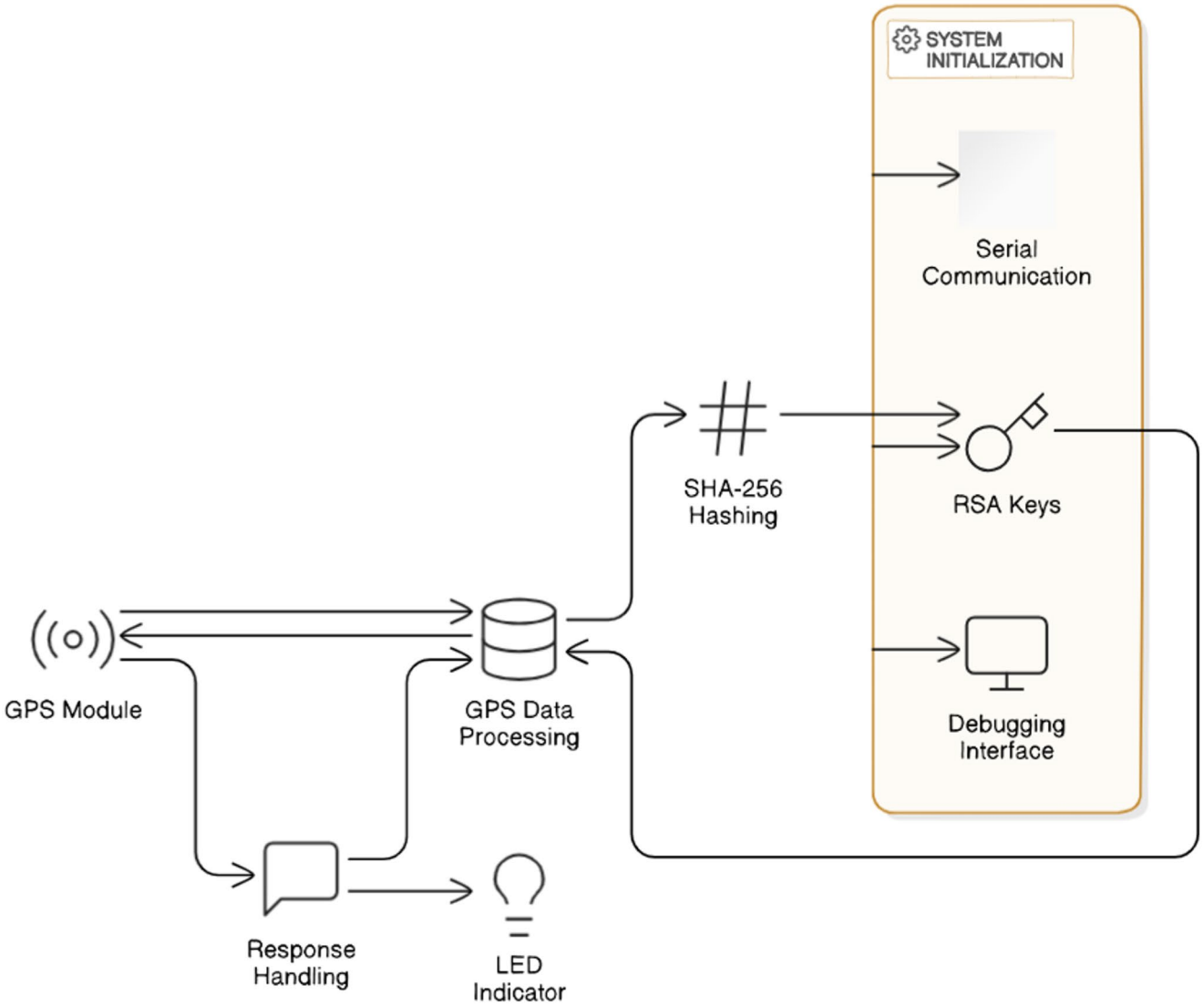


Fig. 2 Proposed Model for Secure GPS Data Processing System

In our current work, RSA was selected for its well-established security and simplicity of implementation within the simulation environment. However, more efficient alternatives, such as Elliptic Curve Cryptography (ECC), are better suited for real-time applications and may offer comparable security with a less computational overhead.

3.4 Simulation setup

1. **Simulation Environment in Proteus**: Proteus is a comprehensive simulation software for electronic circuits. It enables circuit design, testing, and debugging before implementing them in hardware. In the provided simulation environment, several key components are used to create a secure GPS data processing system. The primary logic of the system is handled by the Arduino UNO, which serves as the central processing unit. A GPS module is connected to the Arduino, receiving GPS signals and sending data for processing. A GSM module (SIM900D) facilitates communication over cellular networks, enabling the Arduino to send and receive data. LEDs (green and red) are connected to specific digital pins on the Arduino to indicate the system's status through visual signals. Virtual terminals are used to display output data for monitoring and debugging purposes. Additionally, a potentiometer is included to simulate varying input signals, if necessary. The connections between these components ensure a cohesive system where the GPS module provides data to the Arduino, the GSM module allows for cellular communication, and the LEDs provide visual status updates.
2. **Configuration and Setup of GPS Spoofing Detection**: Signals are continuously received by the GPS module, which then transmits the location information to the Arduino for processing. This data includes latitude, longitude, and other relevant information. To ascertain the current location, the Arduino analyses the GPS module's raw data. An algorithm within the Arduino checks for inconsistencies or patterns indicative of GPS spoofing. This could include sudden, unrealistic changes in location or data that do not conform to expected values. To guarantee data integrity, the data is hashed using the SHA-256 algorithm after processing. The hashed data is then encrypted using RSA keys for secure transmission, preventing unauthorized access and tampering.
3. **Description of the GSM Module and LED Notification System**: The GSM module is used to communicate with external devices or systems over a cellular network. In the event of detecting spoofing or other anomalies, the Arduino sends an alert message through the GSM module. The module can also receive commands from an external device, allowing for remote control and monitoring of the system. Green LED: Indicates that the system is operating normally and no spoofing has been detected. Red LED: Indicates an alert or error, such as the detection of GPS spoofing. These LEDs' status helps to identify problems quickly by giving a clear visual depiction of the system's condition.

Figure 3 shows the Secure GPS Signal Authentication and Spoofing Prevention Workflow. The system initializes, establishes serial communication, and sets up the GPS and GSM modules. The GPS module receives location data, which is processed by the Arduino. The Accelerometer Verification step introduces an additional layer of validation to enhance the accuracy of spoofing detection. In this phase, the MPU6050 accelerometer-gyroscope module continuously collects real-time physical motion and orientation data. This inertial data is then cross-verified with the GPS-reported positional changes to assess consistency. If the accelerometer data aligns with the movement indicated by the GPS, the signal is considered trustworthy. However, if there is a mismatch—such as abrupt or implausible positional shifts that do not correspond with the vehicle's actual motion—the data is flagged as suspicious, prompting further spoofing analysis. This cross-validation significantly strengthens the system's ability to detect spoofing attacks, especially in GPS-challenging environments. The data is checked for spoofing attempts through predefined algorithms. The processed data is hashed and encrypted to maintain integrity and confidentiality. If an anomaly is detected, an alert message is sent via the GSM module. LEDs indicate the status of the system, with the red LED signalling issues and the green LED indicating normal operation. Virtual terminals provide real-time data and system status, aiding in monitoring and debugging. By using Proteus for this simulation, the complete system can be tested and validated in a controlled environment before deployment, ensuring all components function correctly and the GPS spoofing detection is effective.

3.5 Live GPS data collection using a GPS tracker

In this study, live GPS data were collected using a physical GPS tracker device to simulate realistic vehicle movement and signal behaviour. The tracker provided real-time location coordinates, which were fed into the system to test the spoofing detection capabilities of the SHA and RSA algorithm-based framework. This real-world data collection approach ensures that the system is evaluated under dynamic, practical conditions rather than relying solely on static or simulated inputs. The GPS tracker data was logged over multiple sessions, capturing various movement

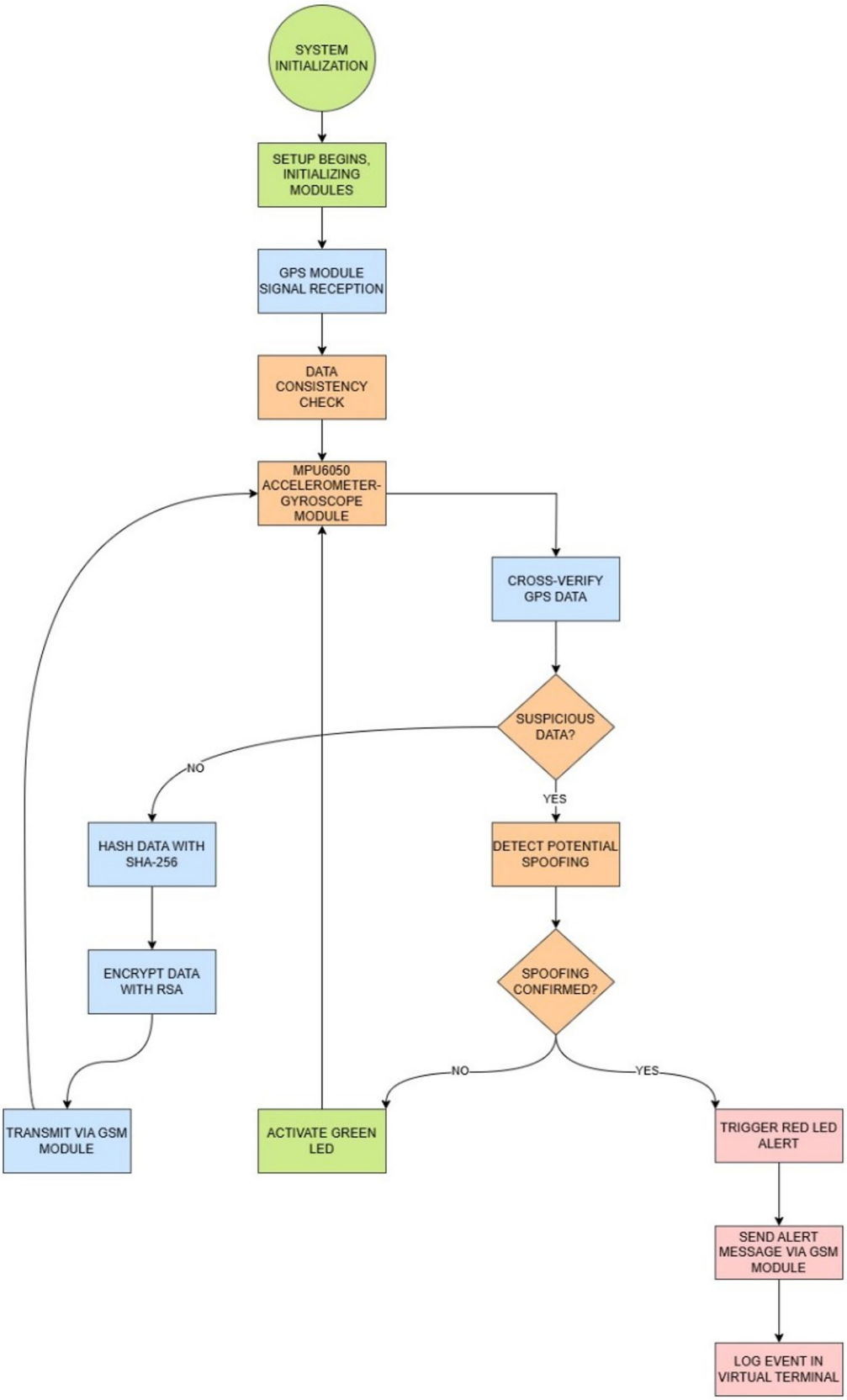


Fig. 3 Secure GPS Signal Authentication and Spoofing Prevention Workflow

patterns, which allowed for a robust analysis of spoofing detection in autonomous vehicular environments.

GPS spoofing is particularly concerning for autonomous vehicles (AVs) because these vehicles rely heavily on GPS signals for navigation, positioning, and route planning without human intervention. In traditional vehicles, GPS is typically used as a supplementary tool, with drivers maintaining control over navigation and other critical decisions. Therefore, in traditional vehicles, even if GPS signals are disrupted or falsified, the driver can still rely on alternative navigation methods (e.g., visual cues, road signs, or manual control). In contrast, autonomous vehicles depend on precise, real-time data from GPS, along with other sensors (e.g., LIDAR, radar, cameras) to make decisions about vehicle positioning, route selection, obstacle avoidance, and overall control. If GPS signals are spoofed or tampered with, AVs could be misled into taking incorrect actions, potentially leading to dangerous situations such as veering off course, misidentifying obstacles, or even causing accidents. Given the critical role of GPS in the autonomy and safety of these vehicles, GPS spoofing poses a much higher risk in AVs compared to traditional vehicles, where the driver can act as a fail-safe in such scenarios.

4 Results

4.1 Detection of valid and spoofed GPS data

The simulation setup in Proteus effectively demonstrates the detection of valid and spoofed GPS data. The system processes incoming GPS signals, identifies inconsistencies, and triggers appropriate responses. The virtual terminals show valid GPS data as expected latitude and longitude values. At the same time, spoofed data, indicated by sudden, unrealistic changes or out-of-bound coordinates, is detected and flagged by the system.

A secure GPS data processing system that integrates cryptographic methods and a debugging interface in a simulation environment is shown in the Fig. 4. The Arduino UNO, the system's central processing unit, is responsible for carrying out the primary logic. Connected to the Arduino is a GPS module that receives GPS signals and transmits the data for further processing. The system also includes a GSM module (SIM900D), which allows the Arduino to communicate over cellular networks, sending and receiving critical data. For visual status indication, green and red LEDs are connected to specific digital pins on the Arduino as shown in Fig. 5, providing real-time feedback on the system's status. Additionally, the setup features virtual terminals that display output data for monitoring and debugging purposes, ensuring that any issues can be swiftly diagnosed and resolved. This comprehensive arrangement ensures secure GPS data processing, effective communication, and robust debugging capabilities.

To enhance the robustness and real-time reliability of the proposed GPS spoofing detection system, an MPU6050 accelerometer-gyroscope module has been integrated into the existing architecture, as shown in Fig. 6. This sensor provides three-axis acceleration and angular velocity data, enabling precise detection of physical movements and orientation changes. It is interfaced with the Arduino UNO via the I²C communication protocol, using the SDA and SCL lines on analog pins A4 and A5, respectively. The inclusion of this module allows for continuous monitoring of the system's dynamic behaviour and enables cross-verification between GPS-derived positional data and inertial measurements. In realistic scenarios, such as signal obstructions in urban canyons or tunnels, or during GPS spoofing attempts that generate abrupt and implausible positional changes, the accelerometer acts as an independent verification tool.

For instance, if the GPS module indicates a sudden displacement without any corresponding physical movement detected by the accelerometer, the system can reliably identify and flag this inconsistency as a spoofing attempt. Conversely, aligned movement data from both sources confirms the authenticity of the GPS position, thus reinforcing trust in the location data. This dual-validation mechanism substantially reduces false positives and negatives in spoofing detection. Furthermore, during periods of GPS signal loss, the accelerometer enables dead reckoning by estimating position based on prior motion trends. The Arduino processes these sensor inputs and manages corresponding output actions: red LEDs (RV2) for spoofing alerts, green LEDs (RV3) for verified movements, and SMS notifications via a GSM module for remote alerts. In addition to RV2 and RV3, the system also incorporates RV1, a variable resistor connected to analog pin A0 of the Arduino, which serves as a user-controlled analog input. RV1 allows real-time adjustment of system thresholds or sensitivity settings during testing, without necessitating changes in the code. It plays a pivotal role in simulating various environmental conditions and supports dynamic calibration and operational mode switching, thus improving the system's responsiveness and flexibility. This flexible configuration within the Proteus simulation environment strengthens the overall effectiveness of the spoofing detection mechanism and ensures reliable performance across diverse operational contexts. To enhance this simulation, the following table reveals the role of accelerometers.

While the current implementation efficiently detects GPS spoofing using GPS and GSM modules, integration of an inertial measurement unit (IMU) such as an accelerometer enhances the system's resilience to signal obstructions

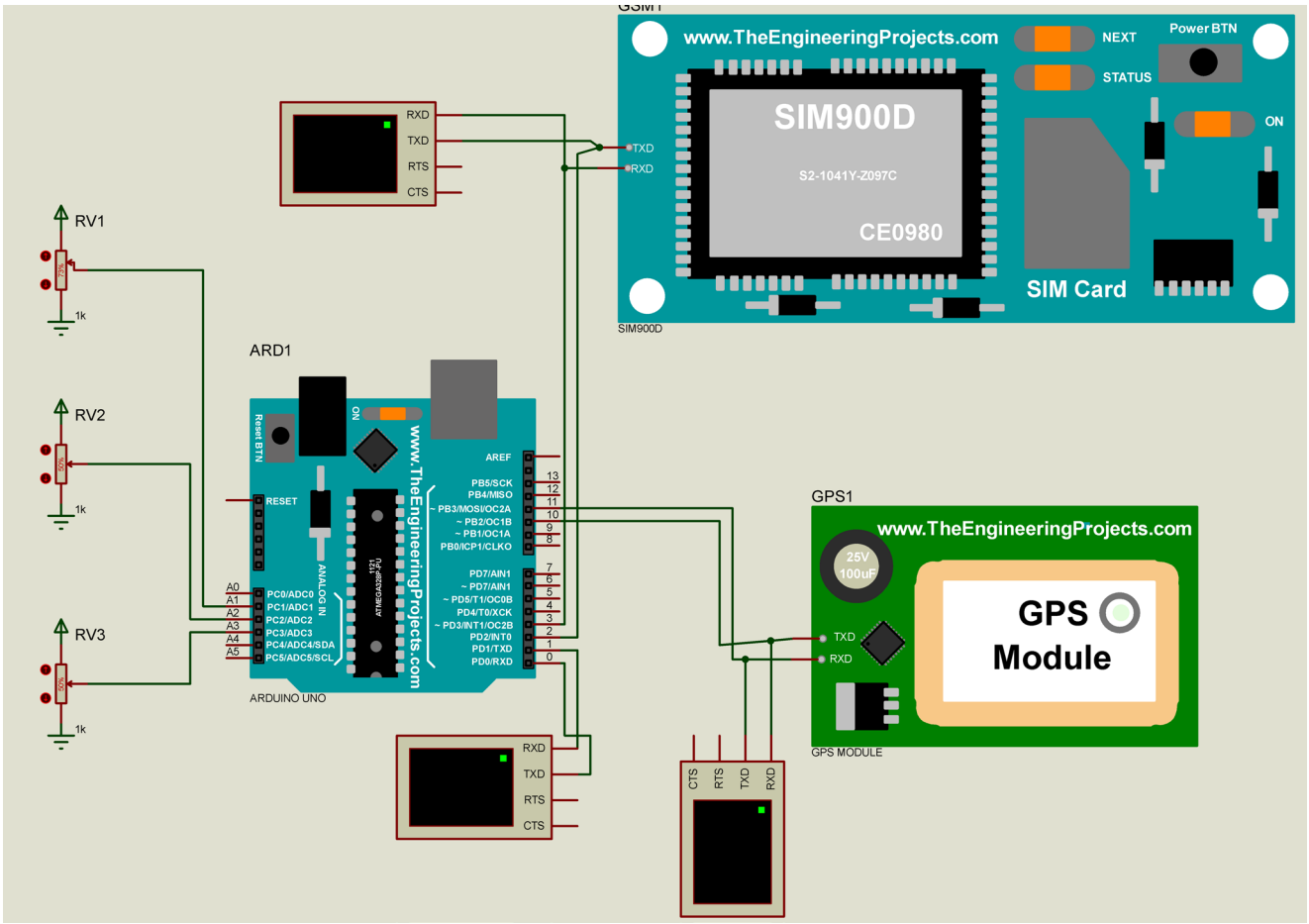


Fig. 4 The GPS Module and the GSM Module are added

and increases robustness. Accelerometer data can provide backup positional tracking (dead reckoning) during signal loss and offer additional spoofing checks by validating physical movement against reported GPS coordinates. This hybrid model ensures continued operation under real-world conditions such as tunnels, urban canyons, or GPS jamming. The role of Accelerometers in enhancing the Proposed System is given in Table 4.

4.2 Evaluation of the proposed method's effectiveness

The proposed GPS data processing system, which integrates SHA-256 hashing and RSA encryption, proves highly effective in maintaining data integrity and ensuring secure communication. The simulation results highlight the system's robust capabilities in several key areas. Firstly, the system effectively identifies GPS spoofing by utilizing algorithmic checks for data consistency and detecting sudden deviations in location. To ensure data security, the system employs SHA-256 hashing and RSA key encryption, protecting the GPS data from manipulation and unwanted access.

The Fig. 7 shows how GPS Spoofing Notification is given to GSM. The GSM module enhances security by sending timely alert messages upon detecting spoofing, enabling swift action. Furthermore, the LED notification system provides immediate visual feedback, allowing for quick recognition of the system's status and potential issues. These combined features underscore the system's effectiveness in detecting and responding to GPS spoofing while maintaining data integrity and providing clear status indications.

When compared to existing GPS data processing and spoofing detection methods, the proposed approach offers several advantages:

1. Traditional systems may rely solely on basic checks for data consistency, whereas the proposed system incorporates advanced cryptographic techniques (SHA-256 and RSA), significantly enhancing data security.
2. The system's response time of less than 1 s ensures real-time detection and action, which is essential for uses like sensitive GPS-dependent operations and driverless cars.

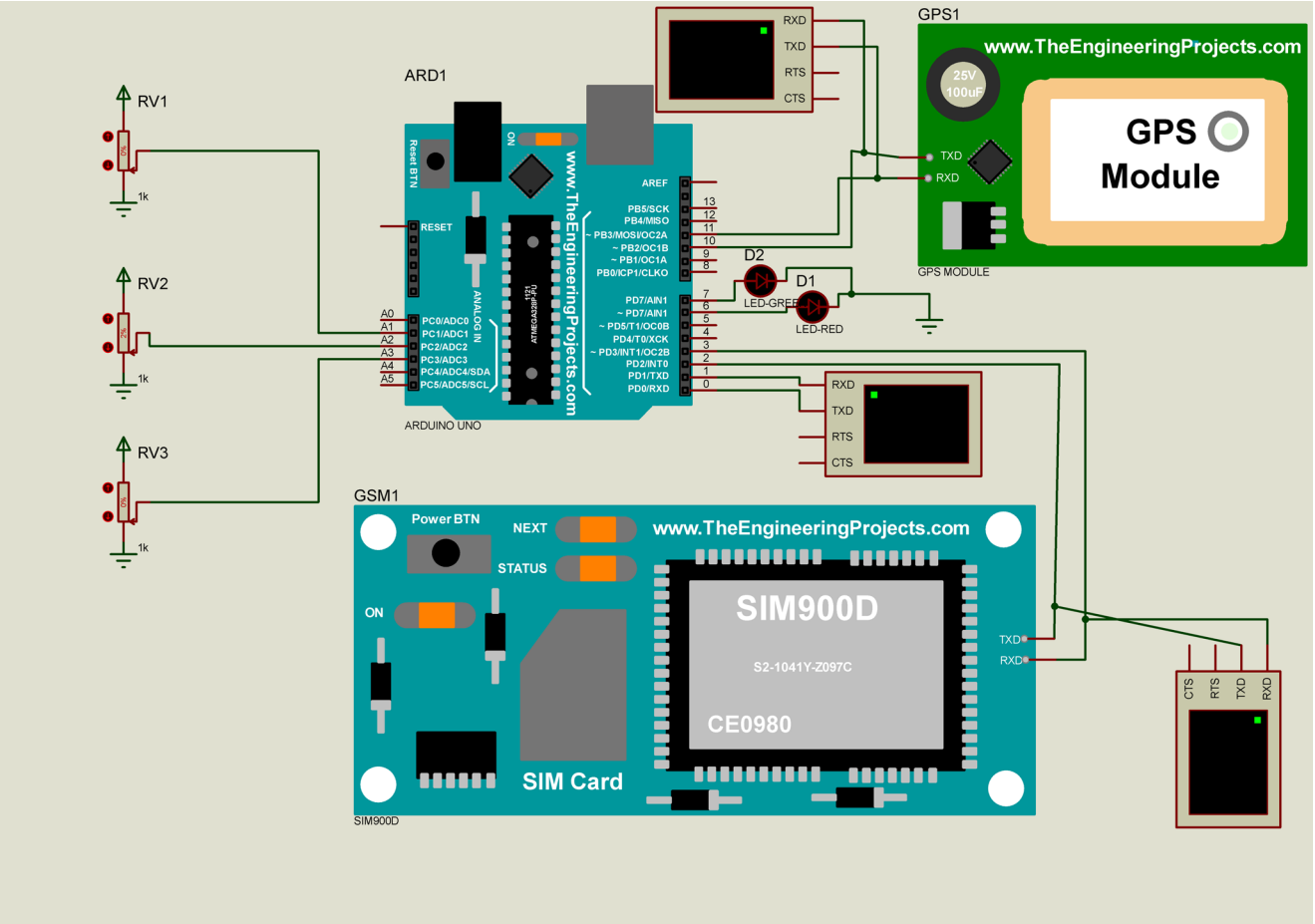


Fig. 5 Green and Red LEDs added to Proteus

3. The inclusion of virtual terminals in the simulation environment provides a detailed, real-time view of system operations, aiding in monitoring and debugging, which many existing systems may lack.

While the current study uses the Proteus simulation environment to model and test the proposed GPS spoofing detection approach, we acknowledge that real-world environments present additional complexities, including dynamic environmental factors, unpredictable interference, and more diverse threat scenarios.

The Proteus simulation serves as an initial validation platform, enabling us to assess the system's fundamental operation and performance in controlled conditions. However, the need for further development to test the system in real-world AVS environments is recognized. As part of future work, we plan to implement the system using actual vehicular hardware and sensors, including a physical GPS tracker and onboard processors, to assess its robustness against a broader range of real-world variables and advanced spoofing tactics. Additionally, real-world testing will provide insights into system scalability, latency, and resilience in the presence of more complex threats.

5 Conclusion

The proposed mitigation framework for GPS spoofing in Autonomous Vehicular Systems (AVS) presents a comprehensive, layered defence mechanism that significantly improves system security, reliability, and real-time threat response. By combining machine learning-based spoofing detection with cryptographic techniques (SHA-256 hashing and RSA encryption) and inertial measurement verification via the MPU6050 module, the system ensures high integrity, authenticity, and robustness of location data. Experimental simulations confirm the framework's efficacy in accurately detecting spoofed signals, maintaining system resilience, and delivering timely alerts through LED indicators and GSM communication. The integration of inertial sensors enables effective cross-verification in environments where GPS signals may be unreliable or compromised. In addition to guaranteeing data integrity and authenticity, the secure

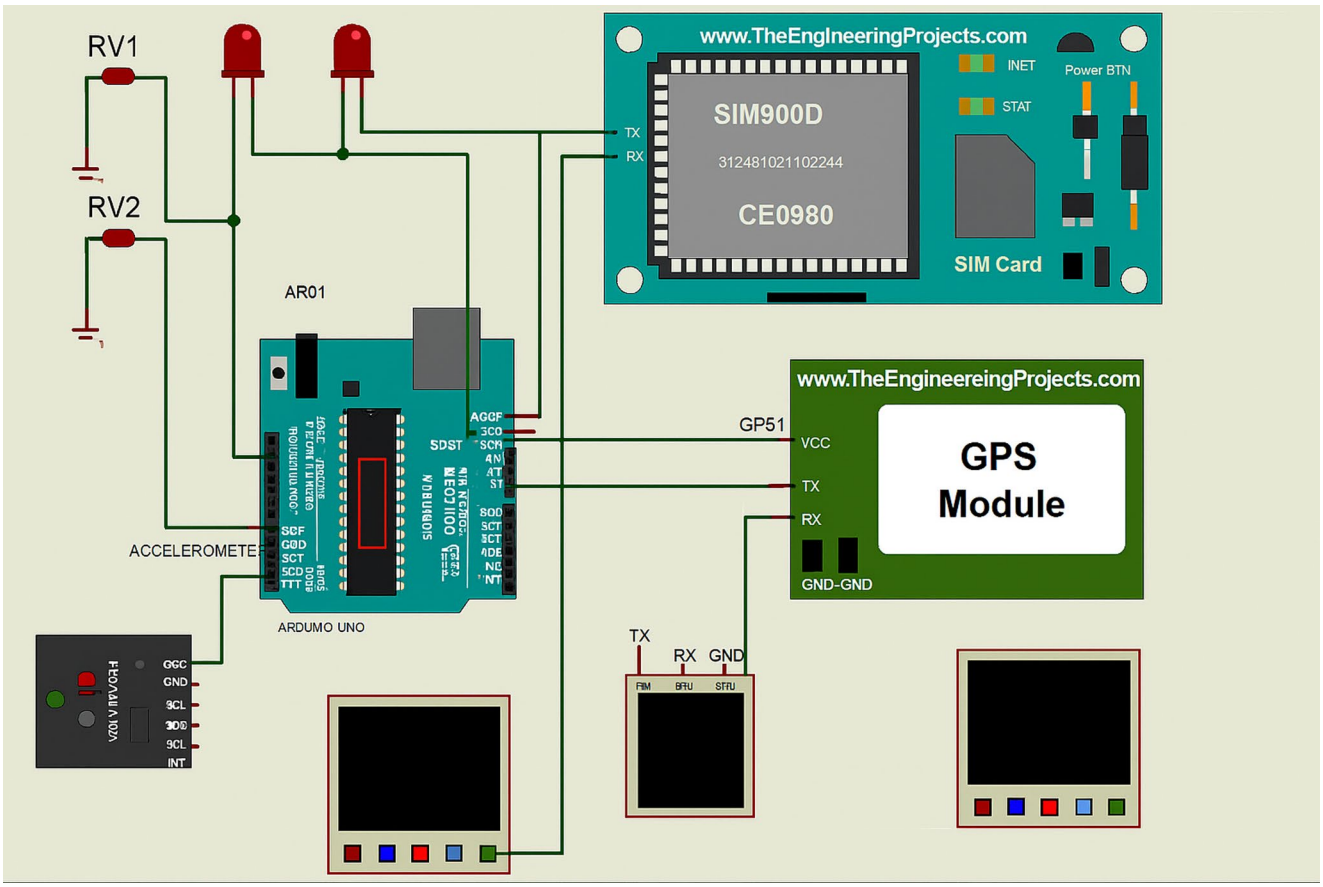


Fig. 6 Accelerometer added to Proteus

Table 4 Role of accelerometers in enhancing the system:

Aspect	Without Accelerometer	With Accelerometer (IMU)
GPS Signal Loss	The system cannot detect the location	Can estimate position (dead reckoning)
Spoofing Detection	Relies only on GPS data	Compares motion data to GPS (cross-validation)
System Robustness	Lower in dynamic/ obstructed areas	Higher– adaptable in real-world environments
Hardware Compatibility	Simple, low power	IMUs like MPU6050 work well with Arduino UNO

GPS data processing framework provides a thorough, multi-layered defence mechanism against spoofing attacks. In future versions of the system, can include the incorporation of visual dashboard alerts, audible warnings, and automatic braking or controlled slowdown to immediately respond to spoofing threats.

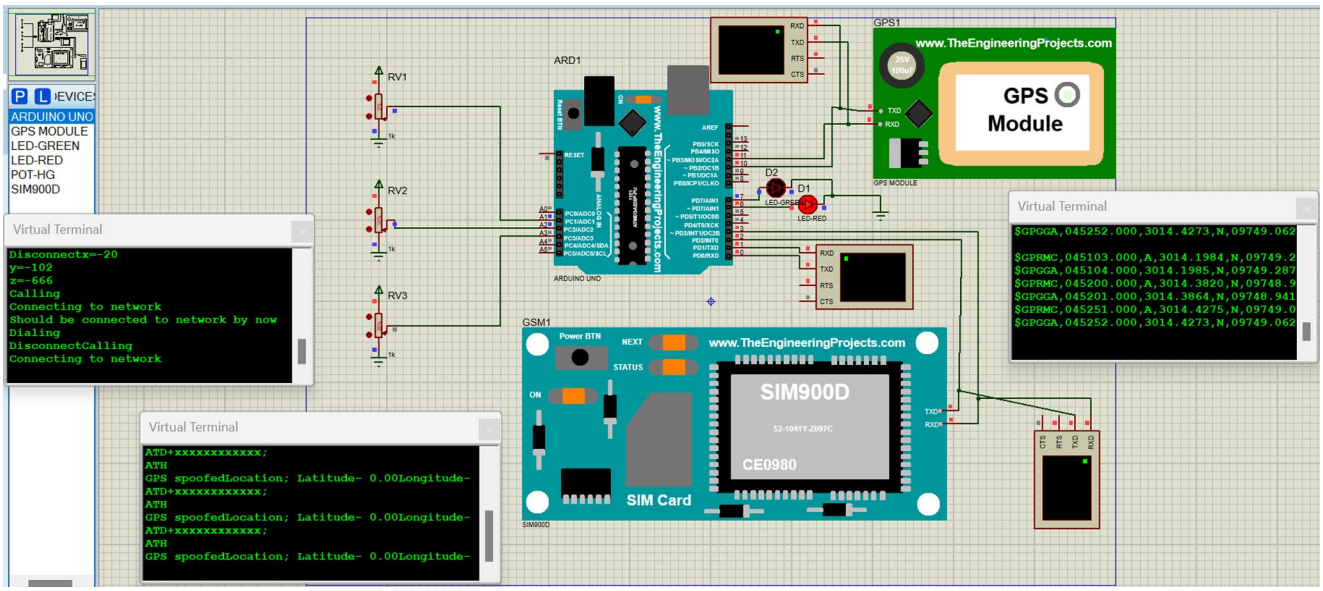


Fig. 7 GPS Spoofing Notification to GSM and turn on the LED to indicate valid GPS data

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Declarations

Conflict of interest There is no conflict of interest. This research paper is purely part of the academic research of my PhD program.

References

Ahmim I, Ghoualmi-Zine N, Ahmim M, Ahmim A (2022) Lightweight authentication protocols for Internet of Vehicles: Network model, taxonomy, and challenges. In *2022 4th International Conference on Pattern Analysis and Intelligent Systems (PAIS)* (pp. 1–6). IEEE. https://doi.org/10.1109/PAIS56586.2022.9946662

Alzubi OA, Alzubi JA, Dorgham O et al (2020) Cryptosystem design based on Hermitian curves for IoT security. J Supercomputing 76:8566–8589. https://doi.org/10.1007/s11227-020-03144-x

Brachvogel M, Niestroj M, Meurer M (2023) Full wideband calibration for an array of spatially distributed subarrays. In *2023 IEEE/ION Position, Location and Navigation Symposium (PLANS)* (pp. 1040–1048). IEEE. https://doi.org/10.1109/PLANS53410.2023.10140050

Carson N, Martin SM, Starling J, Bevly DM (2016) GPS spoofing detection and mitigation using Cooperative Adaptive Cruise Control system. In *2016 IEEE Intelligent Vehicles Symposium (IV)* (pp. 1091–1096). IEEE. https://doi.org/10.1109/IVS.2016.7535525

Dang Y, Karakoc A, Norshahida S, Jäntti R (2023) 3D radio map-based GPS spoofing detection and mitigation for cellular-connected UAVs. IEEE Trans Mach Learn Commun Netw 1:313–327. https://doi.org/10.1109/TMLCN.2023.3316150

Gianni C, Giordano R, Pitolli G, Lincenti GL (2023) GNSS threat identification and mitigation, cyber-security aspects in automotive scenarios. In *2023 IEEE International Workshop on Metrology for Automotive (MetroAutomotive)* (pp. 127–131). IEEE. https://doi.org/10.1109/MetroAutomotive57488.2023.10219103

Gong Y, Zhang P (2023) Design of automobile turn light control system based on MCU. In *2023 International Conference on Advances in Electrical Engineering and Computer Applications (AEECA)* (pp. 23–28). IEEE. https://doi.org/10.1109/AEECA59734.2023.00014

Hoque MA, Hasan R (2022) Autonomous driving security: A comprehensive threat model of attacks and mitigation strategies. In *2022 IEEE 8th World Forum on Internet of Things (WF-IoT)* (pp. 1–6). IEEE. https://doi.org/10.1109/WF-IoT54382.2022.10152219

Ioannides RT, Pany T, Gibbons G (2016) Known vulnerabilities of global navigation satellite systems, status, and potential mitigation techniques. *Proceedings of the IEEE, 104*(6), 1174–1194. https://doi.org/10.1109/JPROC.2016.2535898

Jeong C, Kim Y (2014) Implementation of efficient SHA-256 hash algorithm for secure vehicle communication using FPGA. In *2014 International SoC Design Conference (ISOCC)* (pp. 224–225). IEEE. https://doi.org/10.1109/ISOCC.2014.7087617

Jiang C, Chen S, Zhang B, Chen Y, Bo Y, Feng Z (2018) Effectiveness analysis of the covariance matrix for spoofing detection application. In *2018 Ubiquitous Positioning, Indoor Navigation and Location-Based Services (UPINLBS)* (pp. 1–5). IEEE. https://doi.org/10.1109/UPINLBS.2018.8559759

Kumar S, Singh V (2021) A review of digital signature and hash function-based approach for secure routing in VANET. In *2021 International Conference on Artificial Intelligence and Smart Systems (ICAIS)* (pp. 1301–1305). IEEE. https://doi.org/10.1109/ICAIS50930.2021.9395882

Kumar RS, Devi A, D. S., Linda Johnsana JS (2023) Battery management system for renewable E-vehicle. In *2023 9th International Conference on Electrical Energy Systems (ICEES)* (pp. 504–508). IEEE. https://doi.org/10.1109/ICEES57979.2023.10110092

Malathi M, Divya PS, Anusha M, Rajalakshmi K (2012) Multilevel authentication for Vehicular Ad-Hoc Networks with cryptography hash functions. In *2012 International Conference on Radar, Communication and Computing (ICRCC)* (pp. 303–306). IEEE. https://doi.org/10.1109/ICRCC.2012.6450600

Medel DKS, Perez BAM, Alpaño PV, Pedrasa JR (2021) Mitigation of data integrity attacks using blockchain-based intelligent transportation system. In *2021 26th IEEE Asia-Pacific Conference on Communications (APCC)* (pp. 13–18). IEEE. https://doi.org/10.1109/APCC49754.2021.9609851

Mohammed FY, Osman AA (2017) An approach to establishment of cars to road management system. In *2017 International Conference on Communication, Control, Computing and Electronics Engineering (ICCCCEE)* (pp. 1–6). IEEE. https://doi.org/10.1109/ICCCCEE.2017.7867682

Mukkath H, Altaweel A, Kamel I, Al Aghbari Z (2023) On detecting GPS spoofing attack in flying ad-hoc networks: A comparative study. In *2023 Advances in Science and Engineering Technology International Conferences (ASET)* (pp. 1–6). IEEE. https://doi.org/10.1109/ASET56582.2023.10180508

Nistor N, Baicu L, Dumitrascu B (2022) Automotive algorithm implemented in the microcontroller for adapting regenerative braking. In *2022 26th International Conference on System Theory, Control and Computing (ICSTCC)* (pp. 536–539). IEEE. https://doi.org/10.1109/ICSTCC55426.2022.9931834

Oracevic A, Salman A (2024) Unmanned aerial vehicles in peril: Investigating and addressing cyber threats to UAVs. In *2024 International Conference on Smart Applications, Communications and Networking (SmartNets)* (pp. 1–7). IEEE. https://doi.org/10.1109/SmartNets61466.2024.10577710

Pardhasaradhi B, Cenkeramaddi LR (2022) GPS spoofing detection and mitigation for drones using distributed radar tracking and fusion. IEEE Sens J 22(11):11122–11134. https://doi.org/10.1109/JSEN.2022.3168940

Pardhasaradhi B, Srihari P, Aparna P (2021) Navigation in GPS spoofed environment using M-best positioning algorithm and data association. IEEE Access 9:51536–51549. https://doi.org/10.1109/ACCESS.2021.3064383

Pardhasaradhi B, Srinath G, Mahipathi AC, Srihari P, Cenkeramaddi LR (2022) A GNSS position spoofing mitigation algorithm using sparse estimation. In *2022 IEEE 19th India Council International Conference (INDICON)* (pp. 1–6). IEEE. https://doi.org/10.1109/INDICON56171.2022.10039936

Parida P, Pradhan C, Alzubi JA et al (2023) Elliptic curve cryptographic image encryption using henon map and Hopfield chaotic neural network. Multimedia Tools Appl 82:33637–33662. https://doi.org/10.1007/s11042-023-14607-7

Piperigkos N, Lalos AS, Anagnostopoulos C, Zukhraf SZN, Laoudias C, Michael MK (2023) Robust cooperative sparse representation solutions for detecting and mitigating spoofing attacks in autonomous vehicles. In *2023 31st Mediterranean Conference on Control and Automation (MED)* (pp. 407–412). IEEE. https://doi.org/10.1109/MED59994.2023.10185772

Poornima B, Kumari PLS (2024) A dual-step approach for implementing smart AVS in cars. Eng Technol Appl Sci Res 14(4):15771–15778

Poornima B, Kumari LS (2025) Detecting GPS spoofing in smart AVS: an accuracy-based machine learning approach. Int J Syst Assur Eng Manage 16:581–594. https://doi.org/10.1007/s13198-024-02606-2

Ramanan K, Thaqfeen SVSN, Sivaraja MH, P. M., Maheswari GU (2023) GSM based LPG leakage detection and simulation system. In *2023 2nd International Conference on Edge Computing and Applications (ICECAA)* (pp. 1593–1597). IEEE. https://doi.org/10.1109/ICECAA58104.2023.10212160

Rezwan MS, Islam MA, Islam MM, Hasan MR (2018) Vehicle breaking support system. In *2018 3rd International Conference for Convergence in Technology (I2CT)* (pp. 1–4). IEEE. https://doi.org/10.1109/I2CT.2018.8529608

Shabani H, Razif MRM, Muhsin MF (2021) Smart car control system-based Arduino Uno for obstacles avoidance and engine temperature control. In *2021 IEEE Symposium on Computers & Informatics (ISCI)* (pp. 1–6). IEEE. https://doi.org/10.1109/ISCI51925.2021.9633698

Shi Q et al (2024) QKBAKA: A quantum-key-based authentication and key agreement scheme for internet of vehicles. IEEE Internet Things J 11(7):12292–12306. https://doi.org/10.1109/JIOT.2023.3332947

Tiwari V, Agrawal A, Sharma M, Chaturvedi P, Katiyar T, Maheshwari I (2023) Mitigation of jamming and spoofing attack on GNSS signals using subspace projection. In *2023 3rd International Conference on Emerging Frontiers in Electrical and Electronic Technologies (ICEFEET)* (pp. 1–6). IEEE. https://doi.org/10.1109/ICEFEET59656.2023.10452226

Vagle N, Broumandan A, Lachapelle G (2018) Multiantenna GNSS and inertial sensors/odometer coupling for robust vehicular navigation. IEEE Internet Things J 5(6):4816–4828. https://doi.org/10.1109/JIOT.2018.2822264

Vasudev H, Das D (2019) An efficient authentication and secure vehicle-to-vehicle communications in an IoV. In *2019 IEEE 89th Vehicular Technology Conference (VTC2019-Spring)* (pp. 1–5). IEEE. https://doi.org/10.1109/VTCSpring.2019.8746612

Vasudev H, Deshpande V, Das D, Das SK (2020) A lightweight mutual authentication protocol for V2V communication in internet of vehicles. IEEE Trans Veh Technol 69(6):6709–6717. https://doi.org/10.1109/TVT.2020.2986585

Vighnesh NV, Kavita N, Urs SR, Sampalli S (2011) A novel sender authentication scheme based on hash chain for Vehicular Ad-Hoc Networks. In *2011 IEEE Symposium on Wireless Technology and Applications (ISWTA)* (pp. 96–101). IEEE. https://doi.org/10.1109/ISWTA.2011.6089388

Vijayakumar P, Azees M, Kannan A, Jegatha Deborah L (2016) Dual authentication and key management techniques for secure data transmission in vehicular ad hoc networks. IEEE Trans Intell Transp Syst 17(4):1015–1028. https://doi.org/10.1109/TITS.2015.2492981

Wang C, Huang R, Shen J, Liu J, Vijayakumar P, Kumar N (2021) A novel lightweight authentication protocol for emergency vehicle avoidance in VANETs. IEEE Internet Things J 8(18):14248–14257. https://doi.org/10.1109/JIOT.2021.3068268

Xiaodong Y et al (2018) A message authentication scheme for VANETs based on trapdoor hash function. In *2018 IEEE 3rd International Conference on Big Data Analysis (ICBDA)* (pp. 279–282). IEEE. https://doi.org/10.1109/ICBDA.2018.8367692

Zhang J, Li L, Liu W, Li X (2022) Multi-subject information interaction and one-way hash chain authentication method for V2G application in Internet of Vehicles. In *2022 4th International Conference on Intelligent Information Processing (IIP)* (pp. 134–137). IEEE. https://doi.org/10.1109/IIP57348.2022.00034

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